

```
In [ ]: !git clone https://github.com/NTHung2034/PP-Final_Project.git
        %cd PP-Final_Project
```

```
Cloning into 'PP-Final_Project'...
remote: Enumerating objects: 1211, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (88/88), done.
remote: Total 1211 (delta 67), reused 72 (delta 35), pack-reused 1087 (from 1)
Receiving objects: 100% (1211/1211), 106.25 MiB | 14.92 MiB/s, done.
Resolving deltas: 100% (640/640), done.
/content/PP-Final_Project
```

```
In [ ]: !bash ./scripts/download_cifar10.sh
```

```
Downloading CIFAR-10 dataset...
Data directory: ./data
--2025-12-13 13:15:15-- https://www.cs.toronto.edu/~kriz/cifar-10-binary.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170052171 (162M) [application/x-gzip]
Saving to: 'cifar-10-binary.tar.gz'

cifar-10-binary.tar 100%[=====>] 162.17M  1.74MB/s   in 50s

2025-12-13 13:16:06 (3.21 MB/s) - 'cifar-10-binary.tar.gz' saved [170052171/170052171]

CIFAR-10 dataset downloaded and extracted to ./data
Files:
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_1.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_2.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_3.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_4.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_5.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 test_batch.bin
```

```
In [ ]: !mkdir build
        %cd build
```

```
/content/PP-Final_Project/build
```

```
In [ ]: %cd /content/PP-Final_Project/build
        !cmake .. -DENABLE_CUDA=ON
        # compile
        !make
```

```

/content/PP-Final_Project/build
-- The CXX compiler identification is GNU 11.4.0
-- The C compiler identification is GNU 11.4.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found CUDAToolkit: /usr/local/cuda/targets/x86_64-linux/include (found version "12.5.82")
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- CUDA Toolkit found - GPU support enabled
-- The CUDA compiler identification is NVIDIA 12.5.82 with host compiler GNU 11.4.0
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc - skipped
-- Detecting CUDA compile features
-- Detecting CUDA compile features - done
-- GPU Naive targets: train_gpu_naive, train_extract_naive, test_layers_gpu_naive, test_autoenc
oder_gpu_naive
-- GPU Optimized v1 targets: train_gpu_opt_v1, train_extract_opt_v1
-- GPU Optimized v2 targets: train_gpu_opt_v2, train_extract_opt_v2
-- Configuring done (3.7s)
-- Generating done (0.0s)
-- Build files have been written to: /content/PP-Final_Project/build
[ 1%] Building CXX object CMakeFiles/test_dataloader.dir/tests/test_dataloader.cpp.o
[ 2%] Building CXX object CMakeFiles/test_dataloader.dir/src/data/cifar10_loader.cpp.o
[ 3%] Linking CXX executable test_dataloader
[ 3%] Built target test_dataloader
[ 4%] Building CXX object CMakeFiles/test_layers_cpu.dir/tests/tests_cpu/test_layers.cpp.o
[ 5%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 6%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/relu_cpu.cpp.o
[ 7%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 8%] Building CXX object CMakeFiles/test_layers_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 9%] Linking CXX executable test_layers_cpu
[ 9%] Built target test_layers_cpu
[10%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/tests/tests_cpu/test_autoencod
er.cpp.o
[11%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/conv2d_cpu.cpp.o
[12%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/relu_cpu.cpp.o
[13%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/maxpool_cpu.cpp.o
[14%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/layers/upsample_cpu.cpp.o
[15%] Building CXX object CMakeFiles/test_autoencoder_cpu.dir/src/models/autoencoder_cpu.cpp.o
[16%] Linking CXX executable test_autoencoder_cpu
[16%] Built target test_autoencoder_cpu
[17%] Building CXX object CMakeFiles/train_cpu.dir/src/train_cpu.cpp.o
[18%] Building CXX object CMakeFiles/train_cpu.dir/src/data/cifar10_loader.cpp.o
[19%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/conv2d_cpu.cpp.o
[20%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/relu_cpu.cpp.o
[21%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/maxpool_cpu.cpp.o
[22%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/upsample_cpu.cpp.o
[23%] Building CXX object CMakeFiles/train_cpu.dir/src/models/autoencoder_cpu.cpp.o
[24%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/logger.cpp.o
[25%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/image_utils.cpp.o
[26%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/memory_tracker.cpp.o
[27%] Linking CXX executable train_cpu

```

```
[ 27%] Built target train_cpu
[ 28%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/extract_features_cpu.cpp.o
[ 29%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/data/cifar10_loader.cpp.o
[ 30%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 31%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/relu_cpu.cpp.o
[ 32%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 33%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 34%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 35%] Linking CXX executable extract_features_cpu
[ 35%] Built target extract_features_cpu
[ 36%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/models_svm/train_svm_cpu.cpp.o
[ 37%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/data/cifar10_loader.cpp.o
[ 38%] Building CXX object CMakeFiles/train_svm_cpu.dir/external/svm.cpp.o
[ 39%] Linking CXX executable train_svm_cpu
[ 39%] Built target train_svm_cpu
[ 40%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/train_gpu_naive.cu.o
[ 41%] Building CXX object CMakeFiles/train_gpu_naive.dir/src/data/cifar10_loader.cpp.o
[ 42%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 43%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 44%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 45%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 46%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 47%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 48%] Linking CUDA device code CMakeFiles/train_gpu_naive.dir/cmake_device_link.o
[ 49%] Linking CXX executable train_gpu_naive
[ 49%] Built target train_gpu_naive
[ 50%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models_svm/train_extract_naive.cu.o
[ 51%] Building CXX object CMakeFiles/train_extract_naive.dir/src/data/cifar10_loader.cpp.o
[ 52%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 53%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 54%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 55%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 56%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 57%] Building CUDA object CMakeFiles/train_extract_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 58%] Linking CUDA device code CMakeFiles/train_extract_naive.dir/cmake_device_link.o
[ 59%] Linking CXX executable train_extract_naive
[ 59%] Built target train_extract_naive
[ 60%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/train_gpu_opt_v1.cu.o
[ 61%] Building CXX object CMakeFiles/train_gpu_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 62%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 63%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 64%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 65%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 66%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
```

```

[ 67%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 68%] Linking CUDA device code CMakeFiles/train_gpu_opt_v1.dir/cmake_device_link.o
[ 69%] Linking CXX executable train_gpu_opt_v1
[ 69%] Built target train_gpu_opt_v1
[ 70%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models_svm/train_extract_opt_v1.cu.o
[ 71%] Building CXX object CMakeFiles/train_extract_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 72%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 73%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 74%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 75%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 76%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 77%] Building CUDA object CMakeFiles/train_extract_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 78%] Linking CUDA device code CMakeFiles/train_extract_opt_v1.dir/cmake_device_link.o
[ 79%] Linking CXX executable train_extract_opt_v1
[ 79%] Built target train_extract_opt_v1
[ 80%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/train_gpu_opt_v2.cu.o
[ 81%] Building CXX object CMakeFiles/train_gpu_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 82%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 83%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 84%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 85%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 86%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 87%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 88%] Linking CUDA device code CMakeFiles/train_gpu_opt_v2.dir/cmake_device_link.o
[ 89%] Linking CXX executable train_gpu_opt_v2
[ 89%] Built target train_gpu_opt_v2
[ 90%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models_svm/train_extract_opt_v2.cu.o
[ 91%] Building CXX object CMakeFiles/train_extract_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 92%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 93%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 94%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 95%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 96%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 97%] Building CUDA object CMakeFiles/train_extract_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 98%] Linking CUDA device code CMakeFiles/train_extract_opt_v2.dir/cmake_device_link.o
[100%] Linking CXX executable train_extract_opt_v2
[100%] Built target train_extract_opt_v2

```

```
In [ ]: !./train_gpu_naive
```

CIFAR-10 AUTOENCODER - NAIVE GPU IMPLEMENTATION (PHASE 2)

Using GPU: Tesla T4
Compute Capability: 7.5
Total Global Memory: 15095 MB

Loading CIFAR-10 training data...
Loaded data_batch_1.bin
Loaded data_batch_2.bin
Loaded data_batch_3.bin
Loaded data_batch_4.bin
Loaded data_batch_5.bin
Training data loaded: 50000 images

NAIVE GPU AUTOENCODER TRAINING

Configuration:

Batch size: 64
Epochs: 20
Learning rate: 0.001
Training samples: 50000
Save directory: /content/PP-Final_Project/models/saved_weights_gpu_naive

Batches per epoch: 781
VRAM usage: 308 MB

Epoch [1/20]	Loss: 0.078121	Time: 458.41s	Avg: 458.41s/epoch
Epoch [2/20]	Loss: 0.045858	Time: 461.35s	Avg: 459.88s/epoch
Epoch [3/20]	Loss: 0.040757	Time: 461.33s	Avg: 460.36s/epoch
Epoch [4/20]	Loss: 0.035659	Time: 460.97s	Avg: 460.52s/epoch
Epoch [5/20]	Loss: 0.030961	Time: 461.03s	Avg: 460.62s/epoch
Epoch [6/20]	Loss: 0.027427	Time: 460.95s	Avg: 460.67s/epoch
Epoch [7/20]	Loss: 0.025278	Time: 460.92s	Avg: 460.71s/epoch
Epoch [8/20]	Loss: 0.023885	Time: 461.04s	Avg: 460.75s/epoch
Epoch [9/20]	Loss: 0.022836	Time: 460.82s	Avg: 460.76s/epoch
Epoch [10/20]	Loss: 0.021982	Time: 461.00s	Avg: 460.78s/epoch
Epoch [11/20]	Loss: 0.021251	Time: 460.86s	Avg: 460.79s/epoch
Epoch [12/20]	Loss: 0.020599	Time: 460.80s	Avg: 460.79s/epoch
Epoch [13/20]	Loss: 0.019997	Time: 460.82s	Avg: 460.79s/epoch
Epoch [14/20]	Loss: 0.019430	Time: 460.72s	Avg: 460.79s/epoch
Epoch [15/20]	Loss: 0.018892	Time: 460.40s	Avg: 460.76s/epoch
Epoch [16/20]	Loss: 0.018386	Time: 460.41s	Avg: 460.74s/epoch
Epoch [17/20]	Loss: 0.017907	Time: 460.50s	Avg: 460.73s/epoch
Epoch [18/20]	Loss: 0.017455	Time: 459.97s	Avg: 460.68s/epoch
Epoch [19/20]	Loss: 0.017037	Time: 460.17s	Avg: 460.66s/epoch
Epoch [20/20]	Loss: 0.016653	Time: 460.53s	Avg: 460.65s/epoch

TRAINING COMPLETED

Total training time: 9213.01s
Average time per epoch: 460.65s
Initial loss: 0.078121
Final loss: 0.016653
Loss reduction: 78.683006%

Final VRAM usage: 308 MB

Saving model weights to: /content/PP-Final_Project/models/saved_weights_gpu_naive

Saved conv1 weights: 6912 params, bias: 256 params

Saved conv2 weights: 294912 params, bias: 128 params

Saved conv3 weights: 147456 params, bias: 128 params

Saved conv4 weights: 294912 params, bias: 256 params

Saved conv5 weights: 6912 params, bias: 3 params

Model weights saved successfully!

Training summary saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/training_summary.txt

TRAINING COMPLETED!

In []: !./train_extract_naive

CIFAR-10 FEATURE EXTRACTION - NAIVE GPU (PHASE 4)

=== Step 1: Loading Dataset ===

Loading CIFAR-10 training data...

Loaded data_batch_1.bin

Loaded data_batch_2.bin

Loaded data_batch_3.bin

Loaded data_batch_4.bin

Loaded data_batch_5.bin

Training data loaded: 50000 images

Loading CIFAR-10 test data...

Test data loaded: 10000 images

Training images: 50000

Test images: 10000

=== Step 2: Loading Encoder Weights ===

Encoder weights loaded

=== Step 3: Extracting Features ===

Train batch [50/500]

Train batch [100/500]

Train batch [150/500]

Train batch [200/500]

Train batch [250/500]

Train batch [300/500]

Train batch [350/500]

Train batch [400/500]

Train batch [450/500]

Train batch [500/500]

Test batch [20/100]

Test batch [40/100]

Test batch [60/100]

Test batch [80/100]

Test batch [100/100]

Feature extraction completed: 36.50s

=== Step 4: Saving Features to .bin Files ===

Saved train features: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/train_features.bin

Shape: [50000, 8192]

Saved train labels: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/train_labels.bin

Shape: [50000]

Saved test features: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/test_features.bin

Shape: [10000, 8192]

Saved test labels: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/test_labels.bin

Shape: [10000]

Saved metadata: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/metadata.txt

=== Summary ===

Feature extraction time: 36.50s

All features saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features

FEATURE EXTRACTION COMPLETED!

```
In [ ]: %cd /content/PP-Final_Project/  
!python /content/PP-Final_Project/src/models_svm/cuml_svm_train.py naive
```

/content/PP-Final_Project

```
=====
CIFAR-10 cudaML SVM Training with RBF Kernel [NAIVE]
=====
```

Configuration:

Implementation: NAIVE
Data directory: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features
Kernel: rbf
C: 10.0
Gamma: auto

=== Step 1: Loading Data ===

Loaded features shape: (50000, 8192)
Loaded labels shape: (50000,)
Loaded features shape: (10000, 8192)
Loaded labels shape: (10000,)
Train samples: 50000
Test samples: 10000

Normalizing features...

Features normalized (mean=0, std=1)

=== Step 2: Training SVM with cudaML ===

Initializing SVM classifier...

Training started...

[2025-12-13 15:53:13.541] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:14.392] [CUML] [debug] SMO solver finished after 48 outer iterations, total inner 29506 iterations, and diff 0.000996
[2025-12-13 15:53:14.997] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:15.718] [CUML] [debug] SMO solver finished after 55 outer iterations, total inner 36118 iterations, and diff 0.000995
[2025-12-13 15:53:16.224] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:16.830] [CUML] [debug] SMO solver finished after 46 outer iterations, total inner 28228 iterations, and diff 0.000990
[2025-12-13 15:53:17.335] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:17.964] [CUML] [debug] SMO solver finished after 49 outer iterations, total inner 31833 iterations, and diff 0.000945
[2025-12-13 15:53:18.463] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:18.985] [CUML] [debug] SMO solver finished after 40 outer iterations, total inner 23563 iterations, and diff 0.000983
[2025-12-13 15:53:19.486] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:19.935] [CUML] [debug] SMO solver finished after 35 outer iterations, total inner 19474 iterations, and diff 0.000996
[2025-12-13 15:53:20.435] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:21.020] [CUML] [debug] SMO solver finished after 44 outer iterations, total inner 27486 iterations, and diff 0.001000
[2025-12-13 15:53:21.521] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:22.383] [CUML] [debug] SMO solver finished after 69 outer iterations, total inner 44081 iterations, and diff 0.000994
[2025-12-13 15:53:22.882] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:23.575] [CUML] [debug] SMO solver finished after 54 outer iterations, total inner 32826 iterations, and diff 0.000993
[2025-12-13 15:53:24.075] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:24.560] [CUML] [debug] SMO solver finished after 36 outer iterations, total inner 20114 iterations, and diff 0.000991
[2025-12-13 15:53:25.062] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:25.658] [CUML] [debug] SMO solver finished after 44 outer iterations, total inner 24525 iterations, and diff 0.000981
[2025-12-13 15:53:26.161] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:26.642] [CUML] [debug] SMO solver finished after 36 outer iterations, total inner

nner 20157 iterations, and diff 0.000980
[2025-12-13 15:53:27.144] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:27.615] [CUML] [debug] SMO solver finished after 36 outer iterations, total i
nner 19159 iterations, and diff 0.000984
[2025-12-13 15:53:28.121] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:28.601] [CUML] [debug] SMO solver finished after 35 outer iterations, total i
nner 19426 iterations, and diff 0.000989
[2025-12-13 15:53:29.097] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:29.652] [CUML] [debug] SMO solver finished after 40 outer iterations, total i
nner 22042 iterations, and diff 0.000996
[2025-12-13 15:53:30.145] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:30.822] [CUML] [debug] SMO solver finished after 51 outer iterations, total i
nner 30736 iterations, and diff 0.000973
[2025-12-13 15:53:31.322] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:32.451] [CUML] [debug] SMO solver finished after 88 outer iterations, total i
nner 52117 iterations, and diff 0.000994
[2025-12-13 15:53:32.966] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:34.069] [CUML] [debug] SMO solver finished after 85 outer iterations, total i
nner 53666 iterations, and diff 0.000978
[2025-12-13 15:53:34.566] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:35.645] [CUML] [debug] SMO solver finished after 81 outer iterations, total i
nner 57394 iterations, and diff 0.000990
[2025-12-13 15:53:36.146] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:37.182] [CUML] [debug] SMO solver finished after 81 outer iterations, total i
nner 51907 iterations, and diff 0.000981
[2025-12-13 15:53:37.682] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:38.544] [CUML] [debug] SMO solver finished after 64 outer iterations, total i
nner 43170 iterations, and diff 0.000987
[2025-12-13 15:53:39.043] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:39.935] [CUML] [debug] SMO solver finished after 68 outer iterations, total i
nner 42534 iterations, and diff 0.000986
[2025-12-13 15:53:40.436] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:40.947] [CUML] [debug] SMO solver finished after 39 outer iterations, total i
nner 23535 iterations, and diff 0.000994
[2025-12-13 15:53:41.444] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:41.992] [CUML] [debug] SMO solver finished after 40 outer iterations, total i
nner 23074 iterations, and diff 0.000975
[2025-12-13 15:53:42.493] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:43.501] [CUML] [debug] SMO solver finished after 76 outer iterations, total i
nner 50245 iterations, and diff 0.000975
[2025-12-13 15:53:43.995] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:45.387] [CUML] [debug] SMO solver finished after 106 outer iterations, total
inner 66741 iterations, and diff 0.000997
[2025-12-13 15:53:45.882] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:46.874] [CUML] [debug] SMO solver finished after 75 outer iterations, total i
nner 47647 iterations, and diff 0.000962
[2025-12-13 15:53:47.368] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:48.302] [CUML] [debug] SMO solver finished after 70 outer iterations, total i
nner 43430 iterations, and diff 0.000967
[2025-12-13 15:53:48.810] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:49.364] [CUML] [debug] SMO solver finished after 40 outer iterations, total i
nner 22640 iterations, and diff 0.000996
[2025-12-13 15:53:49.865] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:50.527] [CUML] [debug] SMO solver finished after 49 outer iterations, total i
nner 27235 iterations, and diff 0.000982
[2025-12-13 15:53:51.034] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:51.963] [CUML] [debug] SMO solver finished after 69 outer iterations, total i
nner 45212 iterations, and diff 0.000993
[2025-12-13 15:53:52.466] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:53.381] [CUML] [debug] SMO solver finished after 68 outer iterations, total i
nner 47099 iterations, and diff 0.000952
[2025-12-13 15:53:53.876] [CUML] [debug] Creating working set with 1024 elements

```
[2025-12-13 15:53:54.851] [CUML] [debug] SMO solver finished after 72 outer iterations, total i
nner 48926 iterations, and diff 0.000993
[2025-12-13 15:53:55.345] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:55.837] [CUML] [debug] SMO solver finished after 37 outer iterations, total i
nner 21798 iterations, and diff 0.000965
[2025-12-13 15:53:56.335] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:56.853] [CUML] [debug] SMO solver finished after 38 outer iterations, total i
nner 22195 iterations, and diff 0.000971
[2025-12-13 15:53:57.350] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:58.177] [CUML] [debug] SMO solver finished after 61 outer iterations, total i
nner 39248 iterations, and diff 0.000997
[2025-12-13 15:53:58.677] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:53:59.651] [CUML] [debug] SMO solver finished after 73 outer iterations, total i
nner 44994 iterations, and diff 0.000981
[2025-12-13 15:54:00.150] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:00.614] [CUML] [debug] SMO solver finished after 34 outer iterations, total i
nner 18841 iterations, and diff 0.000999
[2025-12-13 15:54:01.117] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:01.669] [CUML] [debug] SMO solver finished after 39 outer iterations, total i
nner 22011 iterations, and diff 0.000993
[2025-12-13 15:54:02.180] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:02.843] [CUML] [debug] SMO solver finished after 49 outer iterations, total i
nner 30295 iterations, and diff 0.000996
[2025-12-13 15:54:03.347] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:03.731] [CUML] [debug] SMO solver finished after 30 outer iterations, total i
nner 15323 iterations, and diff 0.000973
[2025-12-13 15:54:04.236] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:04.762] [CUML] [debug] SMO solver finished after 39 outer iterations, total i
nner 21332 iterations, and diff 0.000975
[2025-12-13 15:54:05.261] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:05.729] [CUML] [debug] SMO solver finished after 35 outer iterations, total i
nner 19779 iterations, and diff 0.000991
[2025-12-13 15:54:06.224] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:06.872] [CUML] [debug] SMO solver finished after 49 outer iterations, total i
nner 28192 iterations, and diff 0.000997
[2025-12-13 15:54:07.370] [CUML] [debug] Creating working set with 1024 elements
[2025-12-13 15:54:08.064] [CUML] [debug] SMO solver finished after 53 outer iterations, total i
nner 31610 iterations, and diff 0.000988
Training completed in 62.27s
```

=== Step 3: Evaluation ===

Evaluating on training set...

Processed 10000/50000 samples

Processed 20000/50000 samples

Processed 30000/50000 samples

Processed 40000/50000 samples

Processed 50000/50000 samples

Evaluating on test set...

Training Accuracy: 95.98%

Test Accuracy: 63.03%

Confusion matrix saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/confusion_matrix.png

=== Per-Class Accuracy ===

airplane: 70.40%

automobile: 71.60%

bird: 49.00%

cat: 48.20%

deer: 55.80%

dog: 55.30%

frog: 71.70%

horse: 65.70%
ship: 75.20%
truck: 67.40%

=== Classification Report ===

	precision	recall	f1-score	support
airplane	0.6855	0.7040	0.6946	1000
automobile	0.7474	0.7160	0.7314	1000
bird	0.5309	0.4900	0.5096	1000
cat	0.3906	0.4820	0.4315	1000
deer	0.5717	0.5580	0.5648	1000
dog	0.5620	0.5530	0.5575	1000
frog	0.7057	0.7170	0.7113	1000
horse	0.7244	0.6570	0.6890	1000
ship	0.7558	0.7520	0.7539	1000
truck	0.6878	0.6740	0.6808	1000
accuracy			0.6303	10000
macro avg	0.6362	0.6303	0.6324	10000
weighted avg	0.6362	0.6303	0.6324	10000

=== Step 4: Saving Model ===

Model saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/cuml_svm_model.pkl

=====

TRAINING SUMMARY

=====

Configuration:

Implementation: NAIVE
Kernel: rbf
C: 10.0
Gamma: auto

Training time: 62.27s
Training accuracy: 95.98%
Test accuracy: 63.03%

=====

=== Per-Class Accuracy ===

airplane: 70.40%
automobile: 71.60%
bird: 49.00%
cat: 48.20%
deer: 55.80%
dog: 55.30%
frog: 71.70%
horse: 65.70%
ship: 75.20%
truck: 67.40%

=== Classification Report ===

	precision	recall	f1-score	support
airplane	0.6855	0.7040	0.6946	1000
automobile	0.7474	0.7160	0.7314	1000
bird	0.5309	0.4900	0.5096	1000

cat	0.3906	0.4820	0.4315	1000
deer	0.5717	0.5580	0.5648	1000
dog	0.5620	0.5530	0.5575	1000
frog	0.7057	0.7170	0.7113	1000
horse	0.7244	0.6570	0.6890	1000
ship	0.7558	0.7520	0.7539	1000
truck	0.6878	0.6740	0.6808	1000
accuracy			0.6303	10000
macro avg	0.6362	0.6303	0.6324	10000
weighted avg	0.6362	0.6303	0.6324	10000

Training summary saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/training_summary.txt

```
In [ ]: !zip -r models.zip /content/PP-Final_Project/models
        from google.colab import files
        files.download("models.zip")
```

```

adding: content/PP-Final_Project/models/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v1/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv4_weights.bin (deflated
8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/training_summary.txt (deflate
d 43%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv5_weights.bin (deflated
8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv5_bias.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv3_weights.bin (deflated
8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv3_bias.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv1_weights.bin (deflated
8%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/training_summar
y.txt (deflated 65%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/test_labels.bin
(deflated 49%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/test_features.bi
n (deflated 44%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/metadata.txt (de
flated 24%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/confusion_matri
x.png (deflated 16%)
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(deflated 51%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/cuml_svm_model.p
kl (deflated 43%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/svm_features/train_features.b
in (deflated 44%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv4_bias.bin (deflated 2%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv2_bias.bin (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv1_bias.bin (deflated 2%)
adding: content/PP-Final_Project/models/saved_weights_gpu_naive/conv2_weights.bin (deflated
9%)
adding: content/PP-Final_Project/models/saved_weights/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights/epoch_metrics_cpu.txt (deflated 3%)
adding: content/PP-Final_Project/models/saved_weights/svm_results.txt (deflated 47%)
adding: content/PP-Final_Project/models/saved_weights/test_predictions.txt (deflated 65%)
adding: content/PP-Final_Project/models/saved_weights/cpu_final.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_weights.bin (deflated 7%)
adding: content/PP-Final_Project/models/saved_weights/test_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights/train_features_cpu.bin (deflated 50%)
adding: content/PP-Final_Project/models/saved_weights_gpu_opt_v2/ (stored 0%)
adding: content/PP-Final_Project/models/saved_weights_gpu_optimized/ (stored 0%)

```

In []: